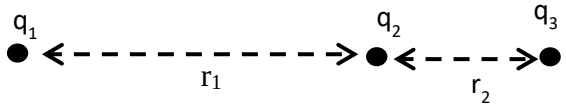
Physics 2 Problem Solving Video # 1 – Electrostatics 1 D

Shown in the figure below, three charged objects are placed along a straight line. The magnitude and type of charge is given below. (k = 9.0 x 10^9 Nm^2/C^2)

q_1 = 5 μC and is positive.
q_2 = 1 μC and is negative.
q_3 = 5 μC and is negative.



r_1 = 0.5 m
r_2 = 0.2 m

A) Determine the net electrostatic force acting on object 1.

Answer: - 0.28 N

B) Determine the net electrostatic force acting on object 2.

Answer: + 0.945 N

C) Determine the net electrostatic force acting on object 3.

Answer: - 0.665 N